

Terminology

15 June 2023 18:20

⑤ terms

1. Random Experiment

An experiment is called random experiment if it satisfies the following two conditions:

- (i) It has more than one possible outcome. ✓
- (ii) It is not possible to predict the outcome in advance ✓

2. Trial →

Trial refers to a single execution of a random experiment. Each trial produces an outcome.

3. Outcome

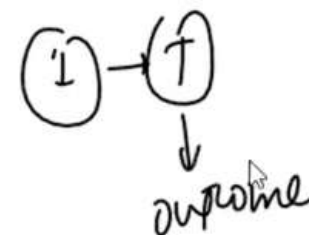
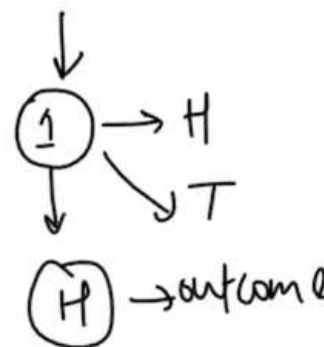
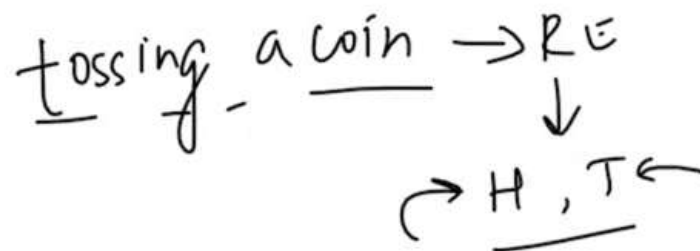
Outcome refers to a single possible result of a trial.

4. Sample Space

Sample Space of a random experiment is the set of all possible outcomes that can occur. Generally, one random experiment will have one set of sample space.

5. Event

Event is a specific set of outcomes from a random experiment or process. Essentially, it's a



4. Sample Space

Sample Space of a random experiment is the set of all possible outcomes that can occur. Generally, one random experiment will have one set of sample space.

5. Event

{ Event is a specific set of outcomes from a random experiment or process. Essentially, it's a subset of the sample space. An event can include a single outcome, or it can include multiple outcomes. One random experiments can have multiple events.



1. Rolling a die
2. Tossing a coin twice
3. Titanic example

- 1) RE $\rightarrow$ rolling a die
- 2) Trial $\rightarrow$ rolling the die once
- 3) Outcome $\rightarrow \{3\}$
- 4) SS $\rightarrow \{1, 2, 3, 4, 5, 6\}$
- 5) Event

- $\rightarrow$ getting a 3 $\rightarrow \{3\}$
- $\rightarrow$ getting a number $> 4 \rightarrow \{5, 6\}$
- $\rightarrow$ getting a number odd $\rightarrow \{1, 3, 5\}$

(1)

RE $\rightarrow$ tossing the coin twice

trial $\rightarrow$ tossing the coin twice (once)
 $\uparrow$

outcome $\rightarrow \{H, T\}$

$\rightarrow$ SS $\rightarrow \{(H, H), (H, T), (T, H), (T, T)\}$

Event $\rightarrow$ getting 2 heads $\{(H, H)\}$

getting at least 1 head $\{(H, H), (H, T), (T, H)\}$

3 outcomes

More videos



↓
Titanic → 891 passengers → Pclass

RE → randomly drawing out a passenger
trial → and finding its Pclass

Outcome → {1}

SS → {1, 2, 3}

Event → A → the passenger is from Pclass = 1 {1}

B → not from Pclass = 1 {2, 3}

More videos



1. Simple Event: Also known as an elementary event, a simple event is an event that consists of exactly one outcome.

For example, when rolling a fair six-sided die, getting a 3 is a simple event.

2. Compound Event: A compound event consists of two or more simple events.

For example, when rolling a die, the event "rolling an odd number" is a compound event because it consists of three simple events: rolling a 1, rolling a 3, or rolling a 5.

3. Independent Events: Two events are independent if the occurrence of one event does not affect the probability of the occurrence of the other event.

For example, if you flip a coin and roll a die, the outcome of the coin flip does not affect the outcome of the die roll.

4. Dependent Events: Events are dependent if the occurrence of one event does affect the probability of the occurrence of the other event.

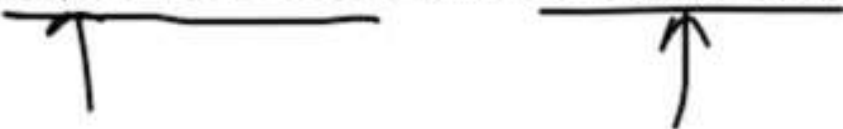
For example, if you draw two cards from a deck without replacement, the outcome of the first draw affects the outcome of the second draw because there are fewer cards left in the deck.

5. Mutually Exclusive Events: Two events are mutually exclusive (or disjoint) if they cannot both occur at the same time.

For example, when rolling a die, the events "roll a 2" and "roll a 4" are mutually exclusive because a single roll of the die cannot result in both a 2 and a 4.

6. Exhaustive Events: A set of events is exhaustive if at least one of the events must occur when the experiment is performed.

For example, when rolling a die, the events "roll an even number" and "roll an odd number" are exhaustive because one or the other must occur on any roll.

7. Impossible event and Certain Event
- 
- The diagram consists of two horizontal lines. The left line is labeled 'Impossible event' and the right line is labeled 'Certain Event'. Below the left line, there is a vertical line with a small horizontal tick at the top, resembling an upward-pointing arrow. Below the right line, there is a vertical line with a small horizontal tick at the top, resembling a downward-pointing arrow. A mouse cursor is visible above the 'Certain Event' label.

Empirical Probability Vs Theoretical Probability

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Empirical Probability -

Empirical probability, also known as experimental probability, is a probability measure that is based on observed data, rather than theoretical assumptions. It's calculated as the ratio of the number of times a particular event occurs to the total number of trials.

- A. Suppose that, in our 100 tosses, we get heads 55 times and tails 45 times. What is the empirical probability of getting a head?
- B. Let's say you have a bag with 50 marbles. Out of these 50 marbles, 20 are red, 15 are blue, and 15 are green. You start to draw marbles one at a time, replacing the marble back into the bag after each draw. After 200 draws, you find that you've drawn a red marble 80 times, a blue marble 70 times, and a green marble 50 times. What is the empirical probability of getting a red marble?



H → 55
Trials → 100

$$P(H) = \frac{55}{100}$$

20 red

$$\frac{80}{200}$$

Theoretical Probability

Theoretical (or classical) probability is used when each outcome in a sample space is equally likely to occur. If we denote an event of interest as Event A, we calculate the theoretical probability of that event as:

Theoretical Probability of Event A = $\frac{\text{Number of Favourable Outcomes (that is, outcomes in Event A)}}{\text{Total Number of Outcomes in the Sample Space}}$

- A. Consider a scenario of tossing a fair coin 3 times. Find the probability of getting exactly 2 heads.
- B. Consider a scenario of rolling 2 dice. What is the probability of getting a sum = 7?

[Random Variable] → misleading
 15 June 2023 18:16 → function

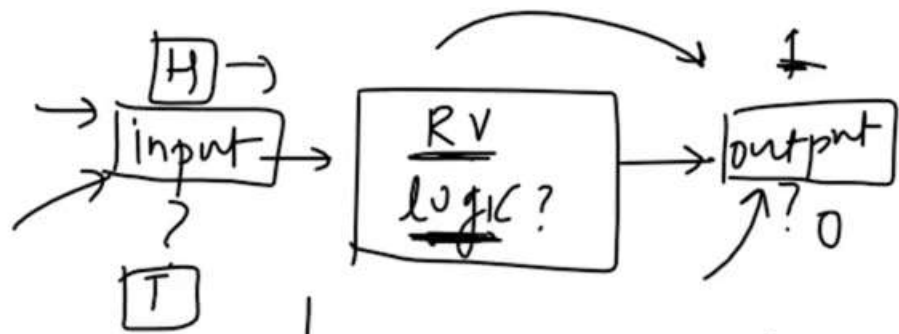
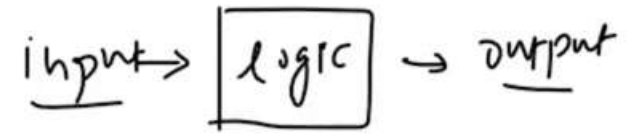
In the context of probability theory, a random variable is a function that maps the outcomes of a random process (known as the sample space) to a set of real numbers.

Input: The input to the function is an outcome from the sample space of a random process.

Output: The output of the function is a real number that we assign to each possible outcome.

Discrete RV
 Continuous RV

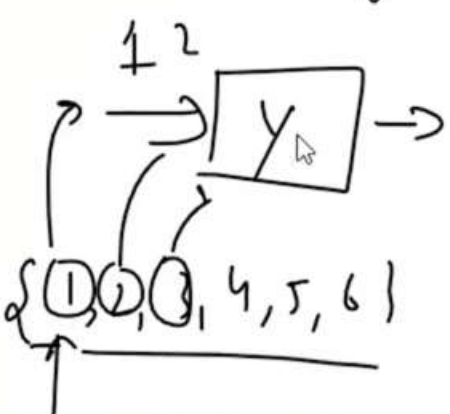
$$X = \left\{ \begin{matrix} 1 & 0 \\ H & T \end{matrix} \right\}$$



random experim
 tossing a coin

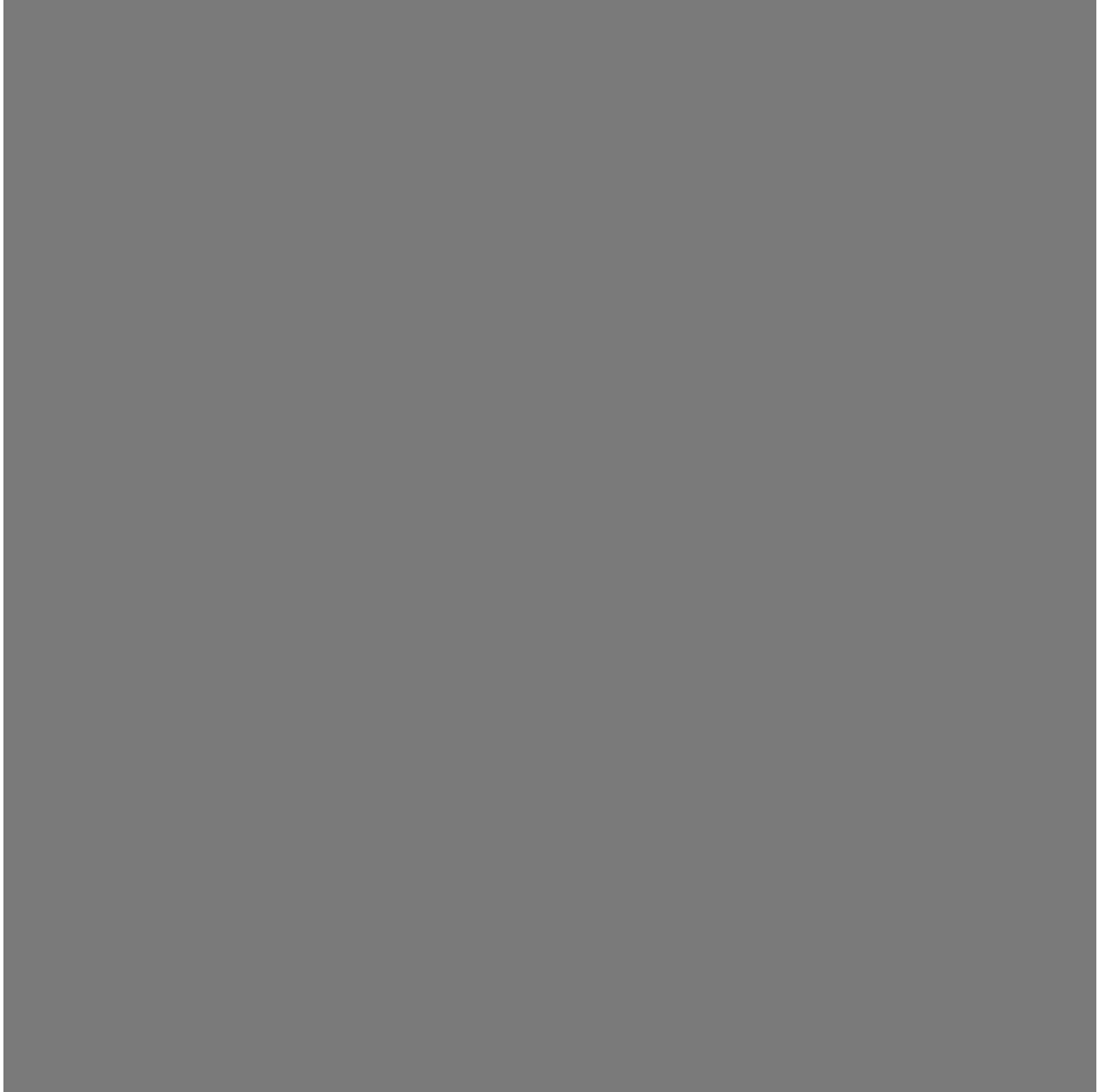
$$\rightarrow SS = \{ \textcircled{H}, \textcircled{T} \}$$

rolling a die



$$Y = \{1, 2, 3, 4, 5, 6\} \text{ [SS]}$$

The transformation from input to output in the function of a random variable is determined by how we choose to define the random variable.
 And the choice of how to define a random variable often depends on the specific aspects of the random process (or event) that you're interested in studying.



Probability Distribution of a Random Variable (Discrete)

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A probability distribution is a list of all of the possible outcomes of a random variable along with their corresponding probability values.

- 1) Toss
- 2) 1 die
- 3) 2 dice

<u>X</u>	1	0
<u>p(X)</u>	$\frac{1}{2}$	$\frac{1}{2}$

Probability dist of
a random var

Toss a coin

$\{H, T\}$

$\rightarrow$ RV $\rightarrow$ X = $\{1, 0\}$

rolling a die
 $S = \{1, 2, 3, 4, 5, 6\}$

$\hookrightarrow$ RV $\rightarrow \{1, 2, 3, 4, 5, 6\}$
 X

X	1	2	3	4	5	6
$P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

probability dist of random variable

sample space

output

(a,b)	1	2	3	4	5	6
1	(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	(6,1)
2	(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2)
3	(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	(6,3)
4	(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	(6,4)
5	(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)
6	(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

rolling 2 dice
 SS = 36 items
 ↓
 RV → sum
 1 2

$P(X=2)$

$X = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

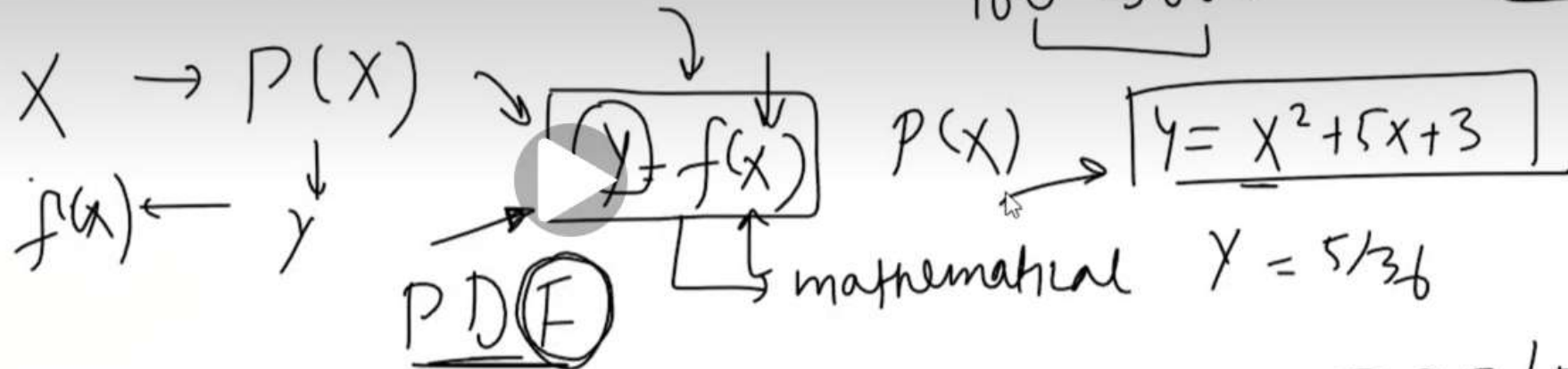
<u>X</u>	2	3	4	5	6	7	8	9	10	11	12
<u>P(X)</u>	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{36}$

Probability Dist Function

100 dice roll

100 → 600

cont
KV



a lot of tiny $\textcircled{X}$

PDF / PMF
↓ ↑
contn discrete
↳ normal

Mean of a Random Variable

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$$\frac{21}{6} = 3.5$$

Mean

The mean of a random variable, often called the expected value is essentially the average outcome of a random process that is repeated many times. More technically, it's a weighted average of the possible outcomes of the random variable, where each outcome is weighted by its probability of occurrence.

$$1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6}$$

rolling a die

X
 $p(X)$

1



3

5

5

6

3.5

$\frac{1}{6}$

$\frac{1}{6}$

$\frac{1}{6}$

$\frac{1}{6}$

$\frac{1}{6}$

$\frac{1}{6}$

Variance of a Random Variable

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The variance of a random variable is a statistical measurement that describes how much individual observations in a group differ from the mean (expected value).

Va

rolling

→ $E[X] = 3.5$

X

,

2

3

4

5

6

normal variance $\rightarrow \underline{\underline{\text{Var}(X)}}$

$$\text{var} = \sum_{i=1}^n \frac{(\underline{x_i} - \underline{\bar{x}})^2}{n-1} =$$

②, 4, 5.

$$V_c$$

$$\text{var} = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1}$$

②, 4, 5, 6, 7, ... 10 dist
 ↑ variance → spread avg

↓
 $(2-3.5)^2$
 $(3-3.5)^2$

↓
 $(X - E[X])^2$
 ↑

↓
 $\text{Var}(X) = E[(X - E[X])^2]$

↓
 mean →  expect

$$\boxed{\text{Var}(X) = E[(X - E[X])^2]}$$

and given random variable X